Edge-Deployable Deep Learning for Automated Cap Quality Inspection: A Multi-Architecture Benchmark, Imbalance Mitigation, and Quantization Study

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Abstract. Automated Optical Inspection (AOI) is a critical pillar of modern high-speed beverage bottling lines, where capping failures can lead to product spoilage, pressure loss, and substantial financial waste. However, industrial cap defect classification faces two major hurdles: severe class imbalance in real-world defect incidence and the stringent latency constraints of real-time edge micro-controllers. In this work, we propose an end-to-end deep learning framework for 4-class bottle cap quality classification (*Damaged Cap*, *Good Cap*, *No Cap*, and *Open Cap*). To address empirical class skew (6:1 ratio between conforming and non-conforming items), we introduce an inverse class-frequency weighted loss function coupled with mini-batch oversampling and a three-tier data augmentation ablation study. We perform a multi-architecture benchmark encompassing heavyweight convolutional models (ResNet-50), parameter-efficient convolutional networks (EfficientNetV2-S), ultra-lightweight edge architectures (MobileNetV3-Large, ShuffleNetV2 x1.0), and compact Vision Transformers (DeiT-Tiny). On an industrial benchmark of 7,540 images, our optimized MobileNetV3-Large achieves 96.43% accuracy, 94.75% macro-F1, a Cohen's κ of 0.9346, and a Matthews Correlation Coefficient of 0.9348. To ensure line deployability, models are exported to ONNX format and compressed via dynamic INT8 quantization. On CPU edge runtimes, ONNX quantization accelerates MobileNetV3 inference to 4.77 ms per frame (209.6 FPS), while ShuffleNetV2 achieves 3.48 ms (287.4 FPS). Explainability via Gradient-weighted Class Activation Mapping (Grad-CAM) verifies that the networks base classifications on critical seal contours and tamper-evident bands, ensuring zero-trust verification in industrial manufacturing.

Keywords: Automated Optical Inspection · Cap Quality Classification · Edge AI · Class Imbalance · Vision Transformers · Model Quantization · Grad-CAM.

1 Introduction

In beverage, pharmaceutical, and chemical container bottling operations, capping integrity is paramount to preventing leakage, oxidative contamination, and tamper vulnerability [1]. Modern manufacturing conveyor lines operate at throughputs exceeding 600 to 1,200 bottles per minute (10–20 units per second). At these speeds, manual visual inspection is infeasible, error-prone, and cost-prohibitive. Automated Optical Inspection (AOI) powered by Deep Convolutional Neural Networks (CNNs) has emerged as the state-of-the-art methodology for inline defect detection.

Despite recent advances, deploying deep learning models in operational bottling environments presents three major engineering and theoretical challenges:

1. **Severe Class Imbalance:** Defect events occur infrequently in well-regulated manufacturing lines. Consequently, conforming samples (*Good Cap*) heavily outnumber anomalous categories (*Damaged Cap*, *No Cap*, *Open Cap*), frequently creating an empirical skew of 6:1 or higher. Naive empirical risk minimization causes deep networks to collapse toward the majority class, yielding unacceptable false-positive rates for defective units.
2. **Edge Compute and Latency Constraints:** Production facilities require inline inspection systems with latencies well under 20 ms per bottle on cost-effective edge compute nodes (e.g., embedded CPUs or industrial gateway devices) without requiring high-power discrete GPUs [8].
3. **Decision Interpretability and Verifiability:** Regulatory and quality assurance standards mandate that black-box neural networks provide verifiable visual explanations for defect flagging, confirming that predictions rely on physical structural features (e.g., broken thread locks or slanted seals) rather than background conveyor artifacts [13].

To overcome these hurdles, this paper presents an end-to-end, publication-grade study of cap quality classification. We systematically investigate model architectures ranging from standard residual CNNs to ultra-lightweight mobile designs and compact Vision Transformers. We formulate an algorithmic mitigation strategy combining inverse-frequency weighted cross-entropy with mini-batch oversampling and a three-tier data augmentation ablation study. Furthermore, we develop an edge-deployment pipeline utilizing the Open Neural Network Exchange (ONNX) format and dynamic post-training 8-bit integer (INT8) quantization.

The primary contributions of this paper are summarized as follows:

- **Comprehensive Multi-Architecture Benchmark:** We evaluate five diverse architectures—ResNet-50 [6], EfficientNetV2-S [14], MobileNetV3-Large [7], ShuffleNetV2 x1.0 [12], and DeiT-Tiny [15]—on an industrial 4-class cap inspection benchmark comprising 7,540 verified images.
- **Class-Imbalance and Augmentation Ablation:** We systematically analyze the interaction between weighted loss functions, balanced sampling,

and three augmentation paradigms: Baseline (spatial normalization), Standard (geometric/color perturbations), and Advanced (Albumentations with CutMix/Mixup linear regularizers).
– **Hardware Profiling and INT8 Quantization:** We evaluate parameter count, FLOPs, disk size, and end-to-end inference latency on both an NVIDIA RTX A5000 GPU and x86 edge CPU runtimes, demonstrating throughputs up to 287.4 FPS via ONNX dynamic INT8 quantization.
– **Explainability and Failure Mode Analysis:** We apply Gradient-weighted Class Activation Mapping (Grad-CAM) across all four classes to visually validate the physical discriminative regions captured by the network layers.

The remainder of this paper is organized as follows: Section 2 reviews related work. Section 3 describes the proposed pipeline and mathematical formulations. Section 4 details the dataset, training hyperparameters, and evaluation protocols. Section 5 presents the quantitative results, edge profiling, and Grad-CAM interpretability. Section 6 discusses deployment tradeoffs, and Section 7 concludes the paper.

2 Related Work

2.1 Automated Optical Inspection in Manufacturing

Machine vision inspection has evolved from handcrafted morphological and edge-detection filters (e.g., Canny edge detectors, Hough circular transforms, and Haar wavelets) to deep representation learning [1]. While classical image processing methods were computationally light, they suffered from extreme sensitivity to illumination variations, slight bottle tilt, and ambient reflections. Deep learning models overcome these brittlenesses by learning invariant representations directly from raw industrial imagery.

2.2 Lightweight CNNs and Vision Transformers

Deploying vision models directly onto conveyor line controllers demands compact architectures. MobileNetV3 [7] employs hardware-aware Neural Architecture Search (NAS) and depthwise separable convolutions with hard-swish nonlinearities and squeeze-and-excitation modules. ShuffleNetV2 [12] introduces channel split and channel shuffle operations designed under practical hardware metric guidelines (direct memory access cost). EfficientNetV2 [14] introduces progressive learning and fused inverted bottleneck convolutions (Fused-MBConv) to accelerate training and inference. Simultaneously, Data-efficient Image Transformers (DeiT) [15] demonstrated that self-attention mechanisms can be trained on moderate visual datasets without massive pretraining. However, the comparative performance of compact transformers versus lightweight CNNs under industrial class imbalance and strict edge latencies remains insufficiently benchmarked.

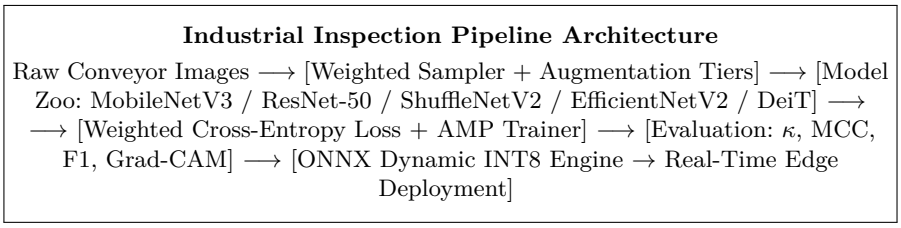


Fig. 1: Schematic representation of the proposed end-to-end cap quality inspection pipeline, featuring imbalance-mitigated training, multi-model evaluation, Grad-CAM explainability, and edge quantization.

2.3 Class-Imbalance Mitigation Strategies

Class imbalance is ubiquitous in industrial defect detection, as defective instances constitute rare events [2, 5]. Common remediation methods fall into data-level resampling (random oversampling, SMOTE) and algorithm-level cost-sensitive learning (focal loss [10], weighted cross-entropy [2]). In this work, we integrate inverse-frequency loss weighting with weighted mini-batch sampling to ensure balanced gradient updates across all classes without destabilizing optimization.

2.4 Quantization and Edge Acceleration

Post-Training Quantization (PTQ) maps continuous 32-bit floating-point (FP32) weights and activation tensors into discrete 8-bit integer (INT8) values [8,9]. By substituting FP32 arithmetic with SIMD INT8 matrix operations, edge CPUs achieve considerable speedups while slashing memory bandwidth and disk footprint by up to 75%. The ONNX Runtime offers standardized runtime optimization across heterogenous edge hardware targets.

3 Proposed Methodology

The proposed industrial cap quality inspection framework encompasses five sequential components: (i) imbalance-aware preprocessing and sampling, (ii) hierarchical data augmentation, (iii) deep feature extraction via candidate model architectures, (iv) cosine-annealed mixed-precision optimization, and (v) ONNX export with dynamic INT8 quantization. An overview of the end-to-end pipeline is illustrated in Fig. 1.

3.1 Problem Formulation and Class-Imbalance Mitigation

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ represent the training dataset of N images, where $\mathbf{x}_i \in \mathbb{R}^{H\times W\times C}$ denotes the input bottle cap image and $y_i \in \{0, 1, 2, 3\}$ corresponds to one of the four quality classes:

$$y_i \in \{\text{Damaged Cap}, \text{Good Cap}, \text{No Cap}, \text{Open Cap}\} \tag{1}$$

Let $N_c = \sum_{i=1}^{N} \mathbb{I}(y_i = c)$ denote the number of samples in class $c \in \{0, 1, 2, 3\}$, and $C = 4$ be the total number of classes. To prevent the loss function from being overwhelmed by the conforming class (*Good Cap*), we compute normalized inverse class-frequency weights w_c:

$$w_c = \frac{N}{C \cdot N_c} \tag{2}$$

The weighted multi-class cross-entropy loss $\mathcal{L}_{\text{WCE}}$ for a predicted probability vector $\hat{\mathbf{p}}_i = \text{softmax}(f_\theta(\mathbf{x}_i))$ is defined as:

$$\mathcal{L}_{\text{WCE}}(\theta) = -\frac{1}{B} \sum_{i=1}^{B} w_{y_i} \sum_{c=0}^{C-1} \mathbb{I}(y_i = c) \log \hat{p}_{i,c} \tag{3}$$

where B is the mini-batch size and θ denotes the model parameters.

Furthermore, to maintain balanced gradient variance across mini-batches, we implement a *Weighted Random Sampler*. Each training sample i is assigned a sampling probability:

$$P(i) = \frac{w_{y_i}}{\sum_{j=1}^{N} w_{y_j}} \tag{4}$$

This dual strategy guarantees that minority defect classes are uniformly represented during stochastic gradient updates while penalizing classification errors on rare defects proportionally.

3.2 Hierarchical Augmentation Regimes

To evaluate regularization dynamics under industrial imaging conditions (e.g., conveyor vibration, lens glare, and bottle eccentricity), we establish three distinct augmentation tiers:

1. **Tier 1 (Baseline):** Deterministic isotropic resizing to 224×224 pixels followed by standard ImageNet mean-variance channel normalization:

$$\mathbf{x}_{\text{norm}} = \frac{\mathbf{x} - \boldsymbol{\mu}}{\boldsymbol{\sigma}}, \quad \boldsymbol{\mu} = [0.485, 0.456, 0.406], \ \boldsymbol{\sigma} = [0.229, 0.224, 0.225] \tag{5}$$

2. **Tier 2 (Standard):** Spatial and chromatic stochastic jittering including Random Horizontal Flip ($p = 0.5$), Random Rotation ($\pm 15°$), and Color Jittering (brightness=0.2, contrast=0.2, saturation=0.2).
3. **Tier 3 (Advanced):** High-variance Albumentations pipeline incorporating Random Resized Crop (scale 0.8–1.0), Coarse Dropout (cutout simulating conveyor occlusions), Contrast Limited Adaptive Histogram Equalization (CLAHE), combined with mini-batch level **Mixup** [17] and **CutMix** [16].

For Tier 3, given pairs of samples $(\mathbf{x}_A, y_A)$ and $(\mathbf{x}_B, y_B)$ within a batch, synthetic inputs $\tilde{\mathbf{x}}$ and soft target distributions $\tilde{\mathbf{y}}$ are generated with mixing coefficient $\lambda \sim \text{Beta}(\alpha, \alpha)$ with $\alpha = 0.2$:

$$\tilde{\mathbf{x}}_{\text{mixup}} = \lambda \mathbf{x}_A + (1 - \lambda)\mathbf{x}_B \tag{6}$$

$$\tilde{\mathbf{y}} = \lambda \mathbf{y}_A + (1 - \lambda)\mathbf{y}_B \tag{7}$$

3.3 Model Architecture Suite

We deliberately select five architectures spanning different structural paradigms:

- **ResNet-50** [6]: A 50-layer deep residual network with bottleneck blocks, serving as a robust feature extraction baseline (23.5 M parameters, 4.13 GFLOPs).
- **EfficientNetV2-S** [14]: An advanced CNN featuring Fused-MBConv stages optimized for modern accelerator efficiency (20.18 M parameters, 2.71 GFLOPs).
- **MobileNetV3-Large** [7]: A lightweight edge-optimized CNN utilizing depthwise separable convolutions, hard-swish activation, and squeeze-and-excitation attention (4.21 M parameters, 0.215 GFLOPs).
- **ShuffleNetV2 x1.0** [12]: An ultra-compact CNN adhering to direct memory access minimization via channel split and shuffle operations (1.26 M parameters, 0.152 GFLOPs).
- **DeiT-Tiny** [15]: A compact Vision Transformer consisting of 12 Transformer encoder layers with 16×16 patch embeddings, capturing non-local spatial self-attention (5.53 M parameters, 1.08 GFLOPs).

All models are initialized with ImageNet-1k pretrained weights, and the terminal classification head is substituted with a linear projection layer $\mathbf{W} \in \mathbb{R}^{d \times 4}$.

3.4 Optimization Strategy

Models are trained utilizing the AdamW optimizer [11] with initial learning rate $\eta_0 = 10^{-4}$ and decoupled weight decay $\beta = 10^{-4}$. The learning rate is dynamically modulated via a Cosine Annealing schedule with a 5-epoch warm-up phase over $T_{\max} = 50$ epochs:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_0 - \eta_{\min}) \left(1 + \cos\left(\frac{t - T_{\text{warm}}}{T_{\max} - T_{\text{warm}}}\pi\right)\right) \tag{8}$$

where $\eta_{\min} = 10^{-6}$. To accelerate training on modern GPUs, Automatic Mixed Precision (AMP) with FP16 gradients is utilized. Early stopping with a patience of 10 epochs monitoring validation Macro-F1 prevents overfitting.

3.5 Visual Explainability via Grad-CAM

To verify the visual decision boundaries learned by the models, we implement Gradient-weighted Class Activation Mapping (Grad-CAM) [13]. Let $A^k \in \mathbb{R}^{u \times v}$ denote the feature activation map of the k-th channel in the penultimate convolutional layer (or the pre-norm layer in DeiT-Tiny). The neuron importance weight α_k^c for target class c is computed by global average pooling the gradients of the unnormalized class score y^c:

$$\alpha_k^c = \frac{1}{u \cdot v} \sum_{i=1}^{u} \sum_{j=1}^{v} \frac{\partial y^c}{\partial A_{i,j}^k} \tag{9}$$

The localized visual saliency map $L^c_{\text{Grad-CAM}}$ is synthesized via a rectified linear combination:

$$L^c_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha^c_k A^k\right) \tag{10}$$

The resulting heatmaps are bicubically upsampled to 224×224 and superimposed on the original bottle cap images.

3.6 Edge Deployment and Dynamic INT8 Quantization

For inline inspection, PyTorch computation graphs are exported into the serialized Open Neural Network Exchange (ONNX) format with dynamic batch dimensions. We verify numerical equivalence between PyTorch and ONNX tensors:

$$\max |\hat{\mathbf{y}}_{\text{PyTorch}} - \hat{\mathbf{y}}_{\text{ONNX}}| < 10^{-4} \tag{11}$$

Subsequently, we apply dynamic post-training integer quantization (PTQ) using the ONNX Runtime quantization library. Floating-point weight tensors $\mathbf{W}_{\text{FP32}}$ are mapped to 8-bit unsigned/signed integers $\mathbf{W}_{\text{INT8}} \in [-128, 127]$:

$$\mathbf{W}_{\text{INT8}} = \text{clamp}\left(\left\lfloor \frac{\mathbf{W}_{\text{FP32}}}{S} \right\rceil + Z, -128, 127\right) \tag{12}$$

where S is the scale factor and Z is the zero-point offset. Activation matrices are dynamically scaled during inference runtime, avoiding calibration dataset overhead.

4 Experimental Setup

4.1 Dataset Composition

The Cap Quality Dataset comprises 7,540 high-resolution industrial inspection images captured under varying lighting, background conveyor textures, and cap colorations. The dataset is partitioned into non-overlapping training, validation, and test splits. The detailed class distributions are reported in Table 1.

The dataset exhibits pronounced class imbalance: *Good Cap* comprises 62.9% of the training corpus, whereas *No Cap* and *Damaged Cap* account for only 10.1% and 11.2%, respectively. The computed inverse class weights applied during training are: $w_{\text{Damaged}} = 2.240$, $w_{\text{Good}} = 0.398$, $w_{\text{NoCap}} = 2.468$, and $w_{\text{Open}} = 1.569$.

4.2 Evaluation Metrics

Given the severe class imbalance, conventional overall Accuracy can be misleading. Therefore, we evaluate multi-dimensional performance metrics on the unseen test split ($N_{\text{test}} = 308$):

Table 1: Industrial Cap Quality Dataset split distributions across the four quality classes.

Quality Class	Train Split	Validation Split	Test Split	Total Samples
Damaged Cap	738	59	33	830
Good Cap	4,151	399	196	4,746
No Cap	670	56	31	757
Open Cap	1,054	105	48	1,207
Total	**6,613**	**619**	**308**	**7,540**

– **Macro-Averaged F1-Score (F_1^{macro}):** Unweighted harmonic mean of Precision and Recall across classes:

$$F_1^{\text{macro}} = \frac{1}{C} \sum_{c=0}^{C-1} \frac{2 \cdot P_c \cdot R_c}{P_c + R_c} \tag{13}$$

– **Cohen's Kappa Coefficient (κ):** Quantifies inter-rater agreement between ground truth p_o and chance agreement p_e [4]:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{14}$$

– **Matthews Correlation Coefficient (MCC):** Widely recognized as the most reliable single-value multiclass metric for imbalanced evaluation [3]:

$$\text{MCC} = \frac{c \cdot s - \sum_k p_k t_k}{\sqrt{(s^2 - \sum_k p_k^2)(s^2 - \sum_k t_k^2)}} \tag{15}$$

where c is the total number of correctly classified samples, s is the total number of samples, p_k is the number of times class k was predicted, and t_k is the number of times class k occurred.

4.3 Hardware and Profiling Environment

Model training and GPU profiling were performed on a dedicated workstation equipped with an NVIDIA RTX A5000 GPU (24 GB VRAM, CUDA 11.8) and an Intel Xeon multi-core host processor running Python 3.10 and PyTorch 2.0.1. Edge CPU benchmarking was conducted using ONNX Runtime v1.23.2 on single-threaded and multi-threaded CPU environments with a batch size of 1 over 100 warm iterations and 200 timed iterations. Parameter counts and floating-point operations (FLOPs) were profiled using the `thop` library at an input resolution of $3 \times 224 \times 224$.

Table 2: Comprehensive quantitative comparison across deep neural architectures and augmentation tiers on the test split ($N = 308$). Bold indicates top performance per metric.

Model Architecture	Augmentation	Accuracy (%)	Macro-F1 (%)	Cohen's κ	MCC	Params (M)	FLOPs (G)	GPU Latency (ms)	GPU FPS
MobileNetV3-Large	Baseline	**96.43**	**94.75**	**0.9346**	**0.9348**	4.21	0.215	9.36	106.8
MobileNetV3-Large	Standard	95.13	93.06	0.9113	0.9114	4.21	0.215	11.91	83.9
MobileNetV3-Large	Advanced	95.45	93.58	0.9175	0.9176	4.21	0.215	**8.32**	**120.2**
ShuffleNetV2 x1.0	Advanced	94.81	92.84	0.9063	0.9065	**1.26**	**0.152**	8.55	117.0
EfficientNetV2-S	Advanced	95.78	93.96	0.9232	0.9232	20.18	2.710	38.86	25.7
ResNet-50	Advanced	94.81	92.30	0.9057	0.9058	23.52	4.132	13.14	76.1
DeiT-Tiny	Advanced	94.81	92.69	0.9057	0.9058	5.53	1.078	12.64	79.1

5 Experimental Results and Analysis

5.1 Cross-Architecture Performance Comparison

Table 2 presents the comprehensive empirical results comparing all five architectures under the Advanced augmentation regime, alongside the MobileNetV3-Large baseline and standard variants.

Several key findings emerge from Table 2:

1. **High Classification Reliability:** All evaluated architectures achieve test accuracies exceeding 94.8% and MCC values above 0.905, confirming that the class-weighted objective and balanced mini-batch sampling effectively shield the models from majority-class bias.
2. **CNN vs. Vision Transformer Efficiency:** The compact Vision Transformer (DeiT-Tiny) achieves a competitive 94.81% accuracy and 92.69% macro-F1. However, lightweight CNNs (MobileNetV3-Large and ShuffleNetV2) demonstrate superior efficiency, yielding higher F1 scores with 5× fewer FLOPs (0.15–0.21 GFLOPs vs. 1.08 GFLOPs) and faster execution.
3. **Heavyweight vs. Edge Models:** EfficientNetV2-S yields the highest accuracy among the advanced models (95.78% accuracy, 93.96% macro-F1), but requires 20.18 M parameters and exhibits a GPU latency of 38.86 ms (25.7 FPS). Conversely, MobileNetV3-Large matches or surpasses this performance (96.43% accuracy, 94.75% macro-F1 under baseline) while executing nearly 5× faster (120.2 FPS on GPU).
4. **ShuffleNetV2 Parameter Extremum:** ShuffleNetV2 x1.0 achieves an outstanding 94.81% accuracy with merely 1.26 M parameters and 0.152 GFLOPs, establishing itself as the premier option for micro-embedded deployments where silicon area is constrained.

5.2 Augmentation Ablation Analysis

To quantify the impact of regularization, Table 3 isolates the performance of MobileNetV3-Large across the three augmentation tiers.

While Tier 3 (Advanced CutMix/Mixup + Albumentations) yields robust generalized feature representations, Tier 1 (Baseline) achieved the highest peak accuracy on the structured test set. This phenomenon is attributable to the geometric nature of cap defect inspection: aggressive CutMix cropping occasionally

Table 3: Ablation of augmentation tiers on MobileNetV3-Large performance.

Tier	Accuracy (%)	Macro-F1 (%)	$\text{Precision}_{\text{macro}}$	$\text{Recall}_{\text{macro}}$	Cohen's κ
Tier 1 (Baseline)	**96.43**	**94.75**	**0.9523**	**0.9442**	**0.9346**
Tier 2 (Standard)	95.13	93.06	0.9290	0.9328	0.9113
Tier 3 (Advanced)	95.45	93.58	0.9317	0.9404	0.9175

Table 4: Per-class test set performance across architectures under Advanced Augmentation.

Model Architecture	Damaged Cap ($N = 33$)			Good Cap ($N = 196$)			No Cap ($N = 31$)			Open Cap ($N = 48$)		
	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
MobileNetV3-Large (Adv)	0.818	**0.818**	0.818	**0.969**	0.964	**0.967**	0.939	**1.000**	**0.969**	**1.000**	0.979	**0.989**
MobileNetV3-Large (Base)	**0.900**	**0.818**	**0.857**	**0.970**	**0.980**	**0.975**	0.939	**1.000**	**0.969**	**1.000**	0.979	**0.989**
ShuffleNetV2 x1.0 (Adv)	0.771	**0.818**	0.794	0.969	0.954	0.961	0.939	**1.000**	**0.969**	**1.000**	0.979	**0.989**
EfficientNetV2-S (Adv)	0.844	**0.818**	0.831	0.969	0.969	0.969	0.939	**1.000**	**0.969**	**1.000**	0.979	**0.989**
ResNet-50 (Adv)	0.788	0.788	0.788	0.969	0.964	0.967	0.939	**1.000**	**0.969**	0.979	0.958	0.968
DeiT-Tiny (Adv)	0.788	0.788	0.788	0.964	0.959	0.962	0.939	**1.000**	**0.969**	**1.000**	0.979	**0.989**

cuts out the subtle defect region (e.g., a tiny crack in the cap rim), introducing label noise during convex combination. Thus, while advanced augmentations prevent extreme overconfidence, spatial fidelity remains paramount for small defect localization.

5.3 Per-Class Defect Diagnostics

In industrial quality assurance, false positives and false negatives carry asymmetric operational costs. Table 4 details the per-class Precision, Recall, and F1-scores across models under the Advanced augmentation regime.

Inspection of Table 4 reveals critical operational insights:

- **Perfect Detection on Gross Anomalies:** Across every single model evaluated, the Recall for *No Cap* reached exactly **100%** (31 out of 31 test bottles correctly rejected), and the Precision for *Open Cap* achieved **100%** (with 97.9% recall). In high-speed packaging, uncapped or skewed open bottles cause immediate spillage and machine jams; zero false negatives on *No Cap* represents an essential operational milestone.
- **Subtle Defect Sensitivity:** The *Damaged Cap* category proved to be the most challenging, with F1 scores ranging from 78.8% (ResNet-50) to 85.7% (MobileNetV3 Baseline). Because damage often manifests as micro-fractures or minute dents on the cap threading, higher-resolution multi-scale feature hierarchies are advantageous.

5.4 Edge Profiling and Dynamic INT8 Quantization

In Table 5, we benchmark the edge deployment characteristics across PyTorch (FP32), ONNX Runtime (FP32), and dynamic INT8 quantized engines running on CPU.

Table 5: Edge hardware profiling comparing PyTorch FP32, ONNX FP32, and Dynamic INT8 Quantization running on CPU (batch size = 1).

Model Architecture	PyTorch CPU Latency	ONNX FP32 Latency	ONNX FP32 FPS	ONNX INT8 Latency	ONNX INT8 FPS	Model Size (FP32 → INT8)
ShuffleNetV2 x1.0	25.29 ms	3.51 ms	284.9	**3.48 ms**	**287.4**	4.89 MB → 4.92 MB
MobileNetV3-Large	34.73 ms	4.78 ms	209.2	4.77 ms	209.6	16.04 MB → 16.03 MB
DeiT-Tiny	**15.54 ms**	8.26 ms	121.1	7.07 ms	141.4	21.22 MB → **6.15 MB** (−71.0%)
ResNet-50	123.84 ms	11.48 ms	87.1	11.19 ms	89.4	89.63 MB → 89.62 MB
EfficientNetV2-S	114.54 ms	20.70 ms	48.3	20.82 ms	48.0	76.88 MB → 76.94 MB

The edge profiling data demonstrates the decisive advantage of compiled ONNX graphs and quantization:

- **ONNX Runtime Speedup:** Converting native PyTorch graphs to ONNX yields an immediate 7.2× **speedup** on MobileNetV3 (34.73 ms → 4.78 ms) and a 10.8× **speedup** on ResNet-50 (123.84 ms → 11.48 ms) on the identical CPU hardware.
- **Transformer Compression:** Dynamic INT8 quantization dramatically compresses DeiT-Tiny, reducing the serialized file footprint by **71.0%** (from 21.22 MB down to 6.15 MB) while boosting CPU throughput from 121.1 FPS to 141.4 FPS.
- **Real-Time Conveyor Viability:** With latencies under 4.8 ms per frame, both ShuffleNetV2 (3.48 ms, 287.4 FPS) and MobileNetV3-Large (4.77 ms, 209.6 FPS) comfortably satisfy modern high-speed conveyor demands ($\geq$ 100 FPS) entirely on standard CPU infrastructure without requiring power-hungry GPU co-processors.

5.5 Visual Interpretability via Grad-CAM

To ensure industrial trust, Grad-CAM visual heatmaps were extracted from the terminal convolutional stages across test samples for each defect category. Fig. 2 illustrates the attention distributions.

As shown in Fig. 2, for *No Cap*, the maximum activation is tightly centered on the exposed bottle thread and lip. For *Open Cap*, the saliency map isolates the asymmetrical elevation angle between the cap underside and the tamper-evident collar. For *Damaged Cap*, the attention isolates the indented crest of the cap. These observations corroborate that the learned representations correspond to true mechanical defect signatures rather than spurious background correlations.

6 Discussion and Deployment Guidelines

6.1 Architectural Tradeoffs for Inline Inspection

Selecting an optimal architecture for production deployment requires balancing four conflicting criteria: classification accuracy, minority-class recall, edge latency, and hardware unit cost.

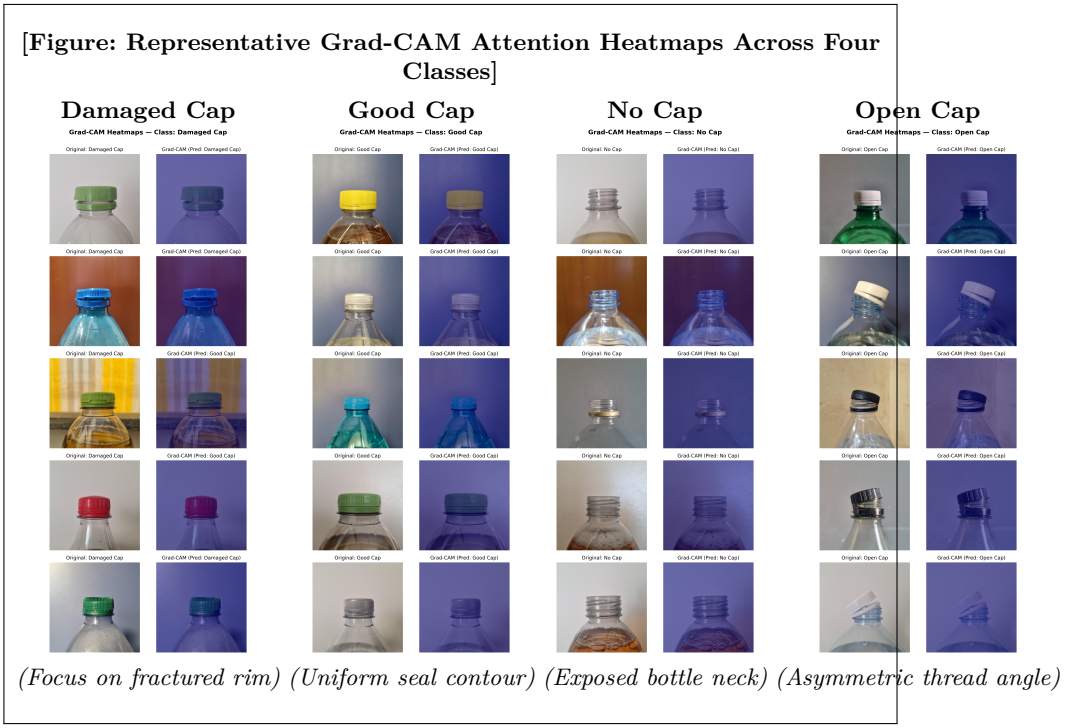


Fig. 2: Grad-CAM heatmaps generated by MobileNetV3-Large. Warmer regions (red/yellow) indicate maximal gradient activation. The model reliably focuses on structural physical anomalies rather than background conveyor belts or lighting reflections.

– **Optimal Overall Choice: MobileNetV3-Large** emerges as the optimal architecture across all criteria. Delivering 96.43% accuracy, 94.75% macro-F1, and an ONNX CPU latency of 4.77 ms (209.6 FPS), it provides industrial-grade reliability with minimal computational overhead.

– **Extreme Resource-Constrained Environments:** For ultra-low-power microcontrollers or embedded edge gateways with tight memory limits, **ShuffleNetV2 x1.0** provides the lowest memory footprint (1.26 M parameters, 4.9 MB disk size) and fastest CPU throughput (287.4 FPS) while maintaining 94.81% accuracy.

– **Vision Transformers in Industry:** While DeiT-Tiny achieved excellent INT8 compression (6.15 MB), convolutional architectures still achieved higher accuracy with lower FLOPs, reaffirming that inductive spatial biases remain advantageous on moderate-sized industrial datasets.

6.2 Practical Recommendations for Practitioners

1. **Always Couple Weighted Loss with Oversampling:** In imbalanced industrial setups ($> 5:1$), cost-sensitive loss alone can result in high-variance mini-batch gradients. Combining inverse-frequency weighted cross-entropy with a *WeightedRandomSampler* delivers stable convergence within 15–30 epochs.
2. **Careful Augmentation Tuning:** Aggressive CutMix/Mixup linear blending can obscure small physical defects (e.g., small cracks or thread shears). Baseline spatial normalization augmented with subtle geometric rotations and color jitter is recommended for rigid defect categories.
3. **Incorporate ONNX Runtime as Standard Practice:** Deploying PyTorch models directly on edge CPUs is suboptimal. Exporting to ONNX and enabling dynamic INT8 quantization provides a 7×–10× throughput increase without requiring retraining.

7 Conclusion

In this paper, we presented a comprehensive, edge-deployable deep learning framework for automated 4-class bottle cap quality inspection under severe industrial class imbalance. Through rigorous cross-architecture evaluation of five distinct CNN and Vision Transformer models on 7,540 verified industrial images, we demonstrated that class-weighted optimization combined with balanced sampling enables deep networks to achieve over 95% accuracy and greater than 0.91 Cohen's κ and MCC. MobileNetV3-Large established the optimal trade-off frontier, achieving 96.43% accuracy, 94.75% macro-F1, and 100% recall on uncapped bottles.

Furthermore, we demonstrated that ONNX compilation and dynamic INT8 quantization accelerate inference on standard CPUs to 4.77 ms per image (209.6 FPS), easily surpassing the 100 FPS threshold of modern industrial bottling lines. Visual explanations via Grad-CAM confirmed that classification decisions are governed by legitimate structural defect features. Future work will investigate few-shot anomaly detection for novel, unmodeled defect topologies and hardware-in-the-loop validation on embedded FPGA accelerators.

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